# Samiur R. Mir

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## Research Interests

Neutrino Physics, Astroparticle Physics, High Energy Astrophysics, Dark Matter

## Education

**Oklahoma State University** — Oklahoma, United States
*Ph.D. in Physics* — *Aug. 2021 – Present*
Advisor: Prof. Vedran Brdar

**University of Alberta** — Alberta, Canada
*M.Sc. in Physics* — *Sept. 2019 – Apr. 2021*

**University of Dhaka** — Dhaka, Bangladesh
*B.Sc. in Physics* — *Jan. 2015 – Aug. 2019*

## Publications & Preprints

7. **S. R. Mir**, Carlos A. Argüelles, and Vedran Brdar, *"New Physics in Ultra-High-Energy Neutrino Flavor"*, arXiv:2611.14624 [hep-ph]

6. Vedran Brdar, **Samiur R. Mir**, Xun-Jie Xu, *"Solar Constraints on Heavy Neutral Leptons with $\nu_\tau$ Mixing"*, arXiv:2608.19319 [hep-ph] (under PRL review)

5. Vedran Brdar, **Samiur R. Mir**, *"Ultra-High-Energy Tau Neutrinos as Probes of Lorentz Invariance"*, JCAP 11 (2026) 008, arXiv:2604.19880 [hep-ph]

4. Vedran Brdar, Dibya S. Chattopadhyay, **Samiur R. Mir**, Tousif Raza, and Marc S. Romanowski, *"Prospects for Neutrino Observation and Mass Measurement from Binary Neutron Star Mergers"*, Phys. Rev. D 113, 123017 (2026), arXiv:2511.16658 [hep-ph]

3. **Samiur R. Mir**, Carlos A. Argüelles, K.S. Babu, Vedran Brdar, *"Double Bangs in IceCube as a Window to the Neutrino Mass Origin"*, Phys. Lett. B 875 (2026) 140322, arXiv:2511.07541 [hep-ph]

2. Vedran Brdar, Ying-Ying Li, **Samiur R. Mir**, Yi-Lin Wang, *"Collider Prospects for the Neutrino Magnetic Moment Portal"*, JHEP 10 (2025) 230, arXiv:2502.07024 [hep-ph]

1. M. Jamil Aslam, Wen Chen, Andrzej Czarnecki, **Samiur R. Mir**, Muhammad Mubasher, *"Rare Decays of the Positronium Ion and Molecule, $\mathrm{Ps}^- \to e^-\gamma$ and $\mathrm{Ps}_2 \to e^+e^-\gamma$, $\gamma\gamma$, $e^+e^-$"*, Phys. Rev. A 104, 052803 (2021), arXiv:2108.06785 [hep-ph]

## Academic Honors and Awards

**Distinguished Graduate Fellowship** | *OSU Foundation and the Graduate College* 2026
**Student Travel Awards** | *granted by APS Fall Meeting* 2025
**Student Travel Awards** | *granted by APS and Pheno organizers* 2024
**Recipient of Honours Scholarship (top 5% CGPA in Bachelors)** | *University of Dhaka* 2019

## Seminar Talks

**Heavy Neutral Leptons from the Sun: New Limits on Mixing with Tau Neutrinos** 2026
Neutrino Theory Community seminar, US (online)

**Heavy Neutral Leptons from the Sun: New Limits on Mixing with Tau Neutrinos** 2026
CNP seminar, Center for Neutrino Physics, Virginia Tech

**RG-Running Neutrino Mixing at IceCube** 2026
Latin American Webinars in Physics (Webinar 195, available on YouTube)

**Probing Neutrinophilic BSM Scenarios with High-Energy Neutrinos** 2026
HEP seminar, PITT PACC, University of Pittsburgh

**Unveiling the Neutrino Magnetic Moment: Constraints from Colliders** 2025
LPPC seminar, Harvard University

## Conference Presentations

**Testing Lorentz Invariance with Ultra-High-Energy Tau Neutrinos** 2026
SUSY Conference, University of Oklahoma

**Testing RG Evolution of Neutrino Mixing with IceCube Double Bang Events** 2026
Pheno Symposium, University of Pittsburgh

**Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube** 2025
Particle Physics on the Plains, University of Kansas

**Probing Neutrino Mass Models with Tau Neutrino Appearance at IceCube** 2025
APS Meeting Texas, Arkansas, Oklahoma Section, Texas Christian University

**Unveiling the Neutrino Magnetic Moment: Constraints from Colliders** 2025
Pheno Symposium, University of Pittsburgh

**Unveiling the Neutrino Magnetic Moment: Constraints from Colliders** 2025
LPPC seminar at Harvard University

**Neutrino Magnetic Moment at Colliders** 2024
Particle Physics on the Plains, University of Kansas

**Collider Probes of Neutrino Magnetic Moment** 2024
DPF - Pheno Symposium, University of Pittsburgh

**Single Photon Decay of the Positronium Ion** 2020
57th Winter Nuclear and Particle Physics Conference, Banff, Alberta

**How to Find Decay Distribution from Symmetry** 2019
10th Annual Symposium for Graduate Physics Research, University of Alberta

## Mentoring Experience

**Safius Sakib Shuddho**, University of Dhaka Aug. 2026–Nov. 2026
ICTP PWF Physics For Bangladesh Internship

**Marc Romanowski**, The College of New Jersey June 2025–Aug. 2025
NSF REU Visiting Undergraduate Researcher

## Summer Schools

**Cosmic Neutrino Practicum** | *Harvard University* June 2026
**International Neutrino Summer School** | *Fermilab* Aug. 2025

## Leadership Activities

**Coordinator and Contributor, HEP-pheno Student Seminar** 2023 – 2025
*Dept. of Physics, Oklahoma State University*
**Vice-President, Physics Graduate Student Association** 2023 – 2025
*Dept. of Physics, Oklahoma State University*
**Outreach Volunteer, Physics Open House** 2023 – 2025
*Dept. of Physics, Oklahoma State University*
**Academic Team Member** | *Bangladesh Mathematical Olympiad* 2015 – 2019
**Academic Instructor** | *Community of Physics, Bangladesh* 2015 – 2019

## Teaching Experience

**Oklahoma State University**
**Teaching Assistant:** Electricity & Magnetism (Labs & Discussions) Aug. 2021 – Present
**University of Alberta**
**Teaching Assistant:** Particles and Waves, Electricity and Magnetism (Labs) Aug. 2019–April 2021

## Professional References

- **Prof. Vedran Brdar** (Ph.D. Supervisor)
  Assistant Professor
  Department of Physics
  Oklahoma State University
  Email: vedran.brdar@okstate.edu
- **Prof. Carlos Argüelles-Delgado**
  Assistant Professor
  Department of Physics
  Harvard University
  Email: carguelles@g.harvard.edu
- **Prof. Kaladi S. Babu**
  Regents Professor of Physics
  Department of Physics
  Oklahoma State University
  Email: kaladi.babu@okstate.edu